

Abhijat Chaturvedi

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Summary

I am a dedicated Machine Learning Engineer with over seven years of experience designing, developing, and deploying advanced machine learning models to solve complex real-world problems.

Experience

WittyBrains

Oct 2025 – Present

Technical Lead

- Leading development of AI-driven chatbots and knowledge assistants using Retrieval-Augmented Generation (RAG) to enhance enterprise workflows.
- Designing scalable vector database architectures with metadata-aware search, embedding pipelines, and incremental deduplication for real-time knowledge retrieval.
- Architecting multimodal LLM systems combining text, structured documents, and images to automate decision-making and improve content generation.
- Implementing tool-using LLM agents with prompt optimization, reliability guardrails, and fallback logic to ensure safe and accurate responses.
- Driving integration of AI microservices into enterprise platforms with best practices in observability, latency reduction, and deployment automation.

Typeface

Jan 2024 – Oct 2025

Machine Learning Engineer

- Solve complex client problems using AI and machine learning techniques.
- Enhance product features by benchmarking against state-of-the-art algorithms.
- Research and implement innovative solutions to improve model performance and efficiency.
- Collaborate with cross-functional teams to integrate AI-driven features into products.

TensorTour (Acquired by Typeface)

Apr 2022 – Dec 2023

Research Engineer

- Developed computer vision models for tasks like image enhancement and super-resolution.
- Designed and managed APIs for deploying machine learning models as services.
- Built and maintained backend infrastructure using Django for AI-driven applications.

CDAC, Noida

Oct 2020 – Apr 2022

Project Engineer

- Developed AI-driven recommendation and decision support systems.
- Designed and implemented chatbot solutions for healthcare and government applications.
- Conducted data analysis and predictive modeling for medical and pandemic-related datasets.

CDAC, Noida

Aug 2018 – Oct 2020

Project Associate

- Built AI-based recommendation systems for investigation processes.
- Developed chatbot solutions to enhance accessibility for nationwide healthcare management.

Projects

Image Quality Metric

- Designing evaluation metrics to assess image quality based on naturalness, object placement, lighting, and edge artifacts.
- Implementing automated methods to quantify realism and detect visual inconsistencies.
- Utilizing deep learning and statistical techniques to enhance evaluation reliability.

Image Harmonization

- Developing models to seamlessly blend objects into generated backgrounds, ensuring realistic visual coherence.
- Addressing inconsistencies in color, lighting, and texture for more natural compositions.
- Leveraging deep learning techniques to improve seamless object integration.

Image Super Resolution

- A comprehensive decision support system for medical practitioners to assist with personal case history analysis, personal medication advice and treatment advice.

Decision Support System

- A comprehensive decision support system for medical practitioners to assist with personal case history analysis, personal medication advice and treatment advice.

Corona Dashboard

- A comprehensive and interactive dashboard on corona statistics in India and the world.
- A corona dashboard for tracking the number of cases, active cases, patients recovered and number of deaths in the first wave of corona in India and the world.
- The project is accessible on GitHub: <https://coronaindia.github.io/>

Publications

- Abhijat Chaturvedi, & Co-Authors. (2020). A Hybrid Approach to Develop and Integrate Chatbot in Health Informatics Systems. *HEALTHINF*.
- Abhijat Chaturvedi, & Co-Authors. (2020). Study and analysis of Unique Health Identifiers and applicability of Aadhaar as a Unique Health Identifier. *2020 IEEE 17th India Council International Conference (INDICON)*.
- Abhijat Chaturvedi, & Co-Authors. (2021). Questionnaire-Based Prediction of Hypertension Using Machine Learning. *Evolving Technologies for Computing, Communication and Smart World: Proceedings of ETCCS 2020*.

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- Abhijat Chaturvedi, & Co-Authors. (2020). Probabilistic Investigation Recommendation System.
- Abhijat Chaturvedi, & Co-Authors. (2020). RaktBot: An AI-Based Citizen Centric Chatbot for Blood Related Services.

Skills

- Machine Learning: Generative AI, LLM, RAG, LangChain, AI Agents
- Frameworks: Pandas, NumPy, Django

Education

Indian Institute of Technology, Jodhpur

M.tech in Artificial Intelligence

Feb 2021 – Dec 2023

Centre for Development of Advanced Computing, Noida

PGDBDA in Big Data Analytics

Feb 2018 – Aug 2018

Greater Noida Institute of Technology, Greater Noida

B.tech in Information Technology

Aug 2013 – July 2017